Roomify Product Requirements Document

Comprehensive Requirements, User Stories & Use Cases

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1. Requirements Gathering

1.1 Stakeholder Analysis & Needs

Stakeholder	Description & Needs
Customers	End users who browse and purchase. **Needs:** Low-friction browsing (guest access), high-fidelity visualization (AR/3D), secure payments, and transparent order tracking.
Administrators	Internal management staff. **Needs:** Real-time analytics (KPIs), efficient product catalog management, delivery personnel oversight, and secure access controls.
Delivery Personnel	Field agents and logistics contractors. **Needs:** Accurate delivery addresses (with map integration), clear item lists, and simple status update tools.

1.2 Functional Requirements

- **Discovery:** Keyword search, category filtering, and price range sliders.
- **Media:** Interactive 3D models (GLB/GLTF) and WebXR/Native AR room overlays.
- **Transaction:** Shopping cart management, secure checkout with multiple payment gateways (Card, COD, Mobile Wallets).
- **Tracking:** Multi-stage status timeline (Placed -> Packed -> Out for Delivery -> Delivered).
- **Management:** Admin KPI dashboard and delivery agent assignment tools.

1.3 Non-functional Requirements

- **Performance:** Fast rendering of 3D assets; polling-based tracking for v1 stability.
- **Security:** Session invalidation on logout; encrypted credentials; role-based access control.
- **Usability:** Toast notifications for feedback; guest access to reduce drop-off.
- **Reliability:** Persistent cart storage; automatic order record creation upon successful payment.

2. User Stories

2.1 Customer Stories

ID	User Story
CS-01	As a visitor, I want to browse all furniture without registering so that I can explore the catalogue before committing.
CS-02	As a visitor, I want to create an account with my name, email, phone, and password so that I can make purchases and track orders. **(Note: v2 includes Phone OTP verification)**.
CS-03	As a registered user, I want to log in using my email or phone number so that I can access my account and cart.
CS-04	As a customer, I want to filter products by category and price so that I can quickly find items that match my needs.
CS-05	As a customer, I want to search by keyword from any page so that I can find specific products instantly.
CS-06	As a customer, I want to see full product details, multiple photos, a 3D view (GLB format), and an AR room preview (WebXR/Native) to make an informed purchase decision.
CS-07	As a customer, I want to add items to a cart and receive a toast confirmation so that I know the item was saved.
CS-08	As a customer, I want to adjust quantities and remove items from my cart so that I can finalise my order before checkout.
CS-09	As a logged-in customer, I want to enter delivery details and choose a payment method (integrated via selected gateway) to complete my purchase.
CS-10	As a customer, I want to see an order confirmation page with my receipt and order number so that I have immediate proof of purchase.
CS-11	As a customer, I want to track my order status and see my delivery person's name via the navbar link.
CS-12	

	As a customer, I want to view/edit my profile and access a settings placeholder (under maintenance) for future features.
CS-13	As a customer, I want to sign out with a confirmation prompt so that I can securely leave my account.

2.2 Admin & Delivery Stories

ID	User Story
AD-01	As an admin, I want to see a dashboard with KPI charts (clicks, conversions, visits) to monitor platform performance.
AD-02	As an admin, I want to add, edit, and delete products (manually in v1) to keep the catalogue accurate.
AD-03	As an admin, I want to view all orders and manually assign delivery personnel to them.
AD-04	As an admin, I want to manage delivery personnel accounts (add/deactivate).
AD-05	As an admin, I want my session to be invalidated on sign-out so that no one can access admin tools without re-authenticating.
DL-01	As a delivery person, I want to log in and see my assigned pending orders and complete history.
DL-02	As a delivery person, I want to see the customer's location, items, and total price for each order so that I can prepare correctly
DL-03	As a delivery person, I want to mark an order as 'on the way' or 'delivered' to update the customer in real-time.
DL-04	As an delivery, I want my session to be invalidated on sign-out so that no one can access delivery tools without re-authenticating.

3. Use Cases

3.1 Customer Use Cases

UC-C01 | Register New Account	
Actor	Visitor (unauthenticated user)
Description	A visitor creates a Roomify account. (Note: serves Egypt/Global regions).
Preconditions	• User is not logged in • User is on the sign-up page
Main Flow	• User clicks 'Sign up' in the navbar • System displays the registration form • User enters: first name, last name, email, phone, password, confirm password • User checks the 'I agree to Privacy Policy and Terms' checkbox • User clicks 'Create account' • System validates all fields (no duplicates, passwords match, valid email/phone format) • System creates the account and starts a logged-in session • System redirects users to the home page with a success notification
Exceptions	• Email or phone already registered — system shows inline error: 'This email/phone is already in use' • Passwords do not match — inline error shown before submission • Required field missing — form prevents submission and highlights empty fields • Privacy checkbox unchecked — form prevents submission with tooltip
Postconditions	• User account is created in the database • User is logged in and on the home page
Future (v2+)	• Social login (Google, Apple) • Phone number OTP verification at registration

UC-C02 | Log In	
Actor	Registered customer
Description	An existing user logs into their Roomify account.
Preconditions	• User has a registered account • User is not currently logged in
Main flow	• User clicks 'Log in' in the navbar • System displays a login modal or page • User enters email or phone number and password • User clicks 'Log in' • System validates credentials • System starts a session and redirects user to the home page

Alt / Exceptions	• Wrong credentials — error: 'Incorrect email/phone or password' • Account does not exist — prompt to register • User accesses cart while logged out — login modal appears before proceeding
Postconditions	• User is authenticated • Session token is stored securely
Future (v2+)	• 'Forgot password' email reset flow • Remember me (persistent sessions)

UC-C03 | Browse & Filter Products	
Actor	Visitor or Logged-in Customer
Description	The user explores the furniture catalogue using category filters and price range.
Main Flow	1. Home page loads. 2. Select category (Sofas, etc.). 3. Adjust price range. 4. System updates grid in real-time.
Exceptions	No items found; network error (retry button).

UC-C04 | Search Products	
Actor	Visitor or logged-in customer
Description	The user searches for specific furniture using a keyword in the navbar search bar.
Preconditions	• User is on any page with the navbar visible
Main flow	• User types a keyword into the search bar in the navbar • User presses Enter or clicks the search icon • System navigates to a search results page • System displays all products matching the keyword with the result count
Alt / Exceptions	• No results found — page shows 'No results for [keyword]' with suggestions • Empty query submitted — search is ignored or shows all products
Postconditions	• User sees a search results page relevant to their query
Future (v2+)	• Live search suggestions as user types (autocomplete dropdown) • Search history

UC-C05 | View Item Detail & Media (AR/3D)	
Actor	Visitor or Customer
Description	The user views full details of a product including photos, 3D model, and AR view.
Preconditions	• User is on the product listing or search results page

Main Flow	• User clicks on a product card • System navigates to the product detail page • Page displays: product name, category tags, price, description, dimensions, colour options, multiple photos • User can click thumbnail images to switch the main photo • User clicks 'View in 3D' — system opens an in-browser 3D model viewer • User clicks 'View in your room (AR)' — system prompts for camera access; AR overlay shown on mobile • User selects colour/variant option • User clicks 'Add to cart' • System adds item to cart and shows a toast notification: 'Added to cart'
Exceptions	• Camera permission denied for AR — system shows a message explaining the requirement • 3D model not available — button is hidden • User not logged in when adding to cart — show modal tell user to login
Postconditions	• Item is in the user's cart • Toast notification confirms the action
Future (v2+)	• Wishlist / save item • Suggested similar items

UC-C06 | Manage Cart	
Actor	Logged-in customer
Description	The user reviews and adjusts their shopping cart before checking out.
Preconditions	• User has at least one item in their cart • User is logged in
Main flow	• User clicks the cart icon in the navbar • System displays the cart page with all saved items • Each item shows: image, name, variant, quantity controls, unit price, line total • User adjusts quantity using + / − controls; line total updates immediately • User clicks 'Remove' to delete an item; item is removed with a toast notification • Cart shows subtotal, delivery fee, and total • User clicks 'Proceed to checkout'
Alt / Exceptions	• User is not logged in when clicking cart — a modal appears with 'Log in' and 'Register' options • Cart is empty — page shows 'Your cart is empty' with a 'Start shopping' link • Item goes out of stock while in cart — warning badge shown on that cart item
Postconditions	• Cart reflects user's final item selection and quantities
Future (v2+)	• Save cart for later • Apply promo/discount codes

UC-C07 | Checkout & Pay	
Actor	Logged-in Customer
Description	Finalizing purchase through selected payment gateway.

Preconditions	• User is logged in • Cart has at least one item • User is on the checkout page
Main Flow	1. Display summary. 2. Enter address. 3. Select payment (Card, COD, Wallet). 4. Process payment. 5. Clear cart; redirect to Confirmation.
Exceptions	• Payment gateway fails — error message shown with a retry option; order is not created • Address fields incomplete — inline validation prevents submission • Session expires mid-checkout — system saves cart and prompts re-login
Postconditions	• Order is created with status 'Pending' • Cart is cleared • User lands on confirmation page

UC-C08 | Order Confirmation	
Actor	Logged-in customer
Description	After a successful payment, the system displays an order confirmation page. This is a v1 addition.
Preconditions	• Payment has been processed successfully
Main flow	• System displays a success icon and 'Order confirmed!' heading • Page shows the order ID (e.g., #FRN-20941) and placement timestamp • Page shows a full receipt: all items, quantities, prices, delivery fee, total paid • Page provides two actions: 'Track my order' and 'Continue shopping' • System also sends an SMS/email notification with the order summary
Alt / Exceptions	• Payment succeeded but confirmation page failed to load — user can access the order via their profile order history [v2]
Postconditions	• User has a record of their order and knows its ID • User can proceed to tracking or continue shopping
Future (v2+)	• Order history page accessible from profile

UC-C09 | Track Order	
Actor	Logged-in Customer
Description	The customer monitors the current status of their order and sees the assigned delivery person.
Preconditions	- User has placed at least one order - Track order' link is in the navbar
Main Flow	1. Click 'Track order' in navbar. 2. Select active order. 3. View timeline (Placed → Packed → Out for Delivery → Delivered). 4. View delivery agent name.
Alt / Exceptions	No active orders — page shows 'No active orders' with a 'Shop now' link Delivery person not yet assigned — field shows 'Pending assignment'

Postconditions	User is informed of their order's current status and estimated delivery

UC-C10 | Manage Profile	
Actor	Logged-in customer
Description	The user views and edits their personal information and manages account security.
Preconditions	• User is logged in
Main flow	• User clicks their avatar/name in the navbar • Dropdown shows three options: Profile, Settings, Sign out • User selects 'Profile' • Profile page shows: full name, email, phone, city • User clicks 'Edit' to enter edit mode and updates any field • User saves changes; system validates and updates the record • User can click 'Change password' — system sends a reset email or prompts for current password • Settings tab shows 'Under maintenance — coming in a future update'
Alt / Exceptions	• New email/phone already in use — inline error • Current password incorrect when changing password — error message
Postconditions	• User's profile data is updated in the database
Future (v2+)	• Profile photo upload • Address book management • App settings

UC-C11 | Sign Out	
Actor	Logged-in customer
Description	The user securely signs out of their account.
Preconditions	• User is logged in
Main flow	• User clicks their avatar in the navbar • User selects 'Sign out' from the dropdown • System displays a confirmation dialog: 'Are you sure you want to sign out?' • User clicks 'Yes, sign out' • System invalidates the session/token • System redirects to the login page
Alt / Exceptions	• User clicks 'Cancel' — dialog closes; user remains logged in
Postconditions	• Session is invalidated • User cannot access protected pages without re-authenticating

3.2 Admin Use Cases

UC-A01 | Admin Dashboard & Analytics	
Actor	Administrator
Description	Admin logs in to view total orders, revenue, unique visitors, and CTR.
Preconditions	• Admin account exists in the system • Admin is on the login page

Main flow	• Admin enters admin email and password • System validates credentials and identifies the admin role • System redirects to the admin dashboard (not the customer home page) • Dashboard displays: total orders, total revenue, unique visitors, click-through rate, conversion rate • A monthly bar chart shows order volume trends
Alt / Exceptions	Wrong credentials — error message; no admin dashboard access
Postconditions	Admin can proceed to manage products/personnel.

UC-A02 | Manage Products	
Actor	Administrator
Description	The admin adds, edits, and removes furniture products from the catalogue.
Preconditions	Admin is logged in and on the Products page
Main flow	• Admin views the products table (name, category, price, stock status) • To add: Admin clicks '+ Add product'; form appears with fields: name, category, description, price, dimensions, colours, images • Admin fills the form and submits; new product appears in the catalogue • To edit: Admin clicks 'Edit' on a product row; pre-filled form appears; admin updates and saves • To delete: Admin clicks 'Delete'; confirmation prompt appears; admin confirms; product is removed
Alt / Exceptions	• Required fields missing on add/edit — form validation prevents submission • Deleting a product that is in an active order — system warns admin and prevents deletion until order is completed
Postconditions	Product catalogue reflects all changes immediately

UC-A03 | Manage Orders	
Actor	Administrator
Description	The admin monitors all orders, views their status, and manages delivery assignments.
Preconditions	• Admin is logged in and on the Orders page
Main flow	• Orders table displays: order ID, customer name, total amount, assigned delivery person, status • Admin can filter orders by status: Pending, On the way, Delivered • Admin can click an order to view full details • Admin can assign or reassign a delivery person to an order
Alt / Exceptions	• No orders exist — empty state shown
Postconditions	• Orders are correctly assigned and their statuses are visible

UC-A04 | Manage Delivery Personnel	
Actor	Administrator
Description	The admin adds, edits, and removes delivery personnel accounts.
Preconditions	• Admin is logged in and on the Deliveries management page
Main flow	• Deliveries table shows: name, phone, orders today, current status (On route / Available / Off duty) • Admin clicks '+ Add person'; form appears with fields: name, phone, zone/area • Admin fills the form and creates the account • Admin clicks 'Manage' to edit a delivery person's details or status

	• Admin can deactivate or remove a delivery person
Alt / Exceptions	• Phone number already registered — duplicate error shown
Postconditions	• Delivery personnel list is accurate • New accounts can log in with their credentials

3.3 Delivery Use Cases

UC-D01 | View & Manage Assigned Orders	
Actor	Delivery person
Description	The delivery person logs in, views their pending and completed deliveries, and updates order statuses.
Preconditions	Delivery person has a registered account created by admin
Main flow	• Delivery person opens the app and logs in with their credentials • System identifies the delivery role and redirects to the delivery home page • Home page shows two sections: 'Pending orders' and 'Delivered today' • Delivery person taps a pending order to open its detail view • Detail view shows: customer address (map preview), list of items, total order value • Delivery person taps 'Mark on the way' when they depart with the order • On arrival and customer receipt, delivery person taps 'Mark delivered' • Order moves from Pending to the Delivered section • Customer's track page reflects the updated status in real time
Alt / Exceptions	• Customer is unavailable at delivery address — delivery person can add a note and attempt redelivery [v2] • Wrong address — delivery person contacts customer via the app [v2]
Postconditions	• Order status is updated • Customer sees the status change on their tracking page • Admin sees updated status in the orders table

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